

Ph.D. Applied Qualifying Exam 2024 (Take Home)

July 4 (9 AM) to July 5 (5 PM)

IMPORTANT - PLEASE READ THESE INSTRUCTIONS CAREFULLY BEFORE YOU START:

This is the Applied Part of the First-Year PhD Qualifying Exam in Statistics. The exam consists of two problems, both of which you need to answer. You are permitted to consult textbooks and other scholarly resources to help you in completing the assignment, but you **may not consult or discuss these problems in any way with anyone during the time that the exam is being administered**. If you draw on information from any outside resources (i.e., resources other than class material and notes) then you should clearly cite the source in your write-up.

You should complete the exam alone, in a room separate from any other test-takers. If you have questions, please email Prof Zhiqiang Tan (ztan@stat.rutgers.edu) for problem 1 and Prof. Han Xiao (hxiao@stat.rutgers.edu) for problem 2, cc'ing Prof Tirthankar Dasgupta (td370@stat.rutgers.edu).

Submission Instructions:

1. Create a folder containing the following:
 - (a) A word or pdf file containing your answers to the two problems, including all tables, figures and Appendices (this may be generated from an R markdown file).
 - (b) An R (or python) file containing the codes used to generate your results, or an R markdown file from which your output html or pdf is generated.
 - (c) A word or a text file that gives a summary of the codes submitted (not required if you submit an R markdown file).
2. Name the folder **yourlastname_yourfirstname_Appliedexam2024**. Zip this folder and email it to hxiao@stat.rutgers.edu AND td370@stat.rutgers.edu by 5 PM July 5, Friday.

Problem 1

Suppose that $X_1, \dots, X_n$ are independent and identically distributed from a p -variate normal distribution with mean vector 0 and unknown variance matrix Σ (assumed to be non-singular). For a fixed vector $\xi^* \in \mathbb{R}^p$, it is of interest to estimate $u^* = \Omega \xi^*$, where $\Omega = \Sigma^{-1}$ is also called the precision matrix.

For p much less than n , a simple estimator of u^* is $\hat{u} = \hat{\Sigma}^{-1} \xi^*$, where $\hat{\Sigma} = \mathbf{X}^T \mathbf{X} / n$, and $\mathbf{X}$ is the $n \times p$ matrix, $\mathbf{X} = (X_1, \dots, X_n)^T$. Equivalently, $\hat{u}$ can be determined as

$$\hat{u} = \operatorname{argmin}_{u \in \mathbb{R}^p} \left\{ \frac{1}{2} u^T \hat{\Sigma} u - u^T \xi^* \right\}.$$

For p close to or larger than n , we consider a Lasso penalized estimator, defined as

$$\hat{u}_\lambda = \operatorname{argmin}_{u \in \mathbb{R}^p} \left\{ \frac{1}{2} u^T \hat{\Sigma} u - u^T \xi^* + \lambda \|u\|_1 \right\}, \quad (1)$$

where $\|\cdot\|_1$ denotes the L_1 norm and $\lambda > 0$ is a tuning parameter. The Lasso penalized estimator is expected to perform well in the setting where the true value u^* is sparse even with p larger than n . In this problem, you are asked to work out numerical algorithms for computing $\hat{u}_\lambda$ for a pre-specified λ .

- 1) For fixed $\lambda > 0$, what is the stationary condition satisfied by $\hat{u}_\lambda$ as a minimizer in (1)?
- 2) Derive a coordinate descent algorithm for computing $\hat{u}_\lambda$. Specifically, denote as $u^0 = (u_1^0, \dots, u_p^0)^T$ the current estimate/iterate. Provide the formula for updating the j th component of u^0 while fixing the remaining components for any j .
- 3) Derive a proximal gradient algorithm for computing $\hat{u}_\lambda$. Specifically, provide the formula for updating the current iterate u^0 to the next, depending on a step size τ .

For a numerical experiment, generate the following data in R (please run these codes exactly with the random seed):

```
#-----  
library(MASS)  
set.seed(1)  
  
### training data  
n <- 600  
p <- 600
```

```

# generate Sigma
rho <- 0.3
Sigma <- matrix(rep(0.3, p^2), nrow = p, ncol = p)
diag(Sigma) <- 1

# generate matrix X and vector xi0 (xi_star in the above description)
X <- mvrnorm(n, rep(0,p), Sigma = Sigma)
xi0 <- Sigma %*% c(-1,1,-1,1,1,rep(0, p-5))
#-----

```

For all questions requiring coding, please provide both your R codes and output. Set the initial value to 0 (a zero vector). The termination criterion for iterations is when the L_1 norm of the difference between current and updated iterates is less than 10^{-5} .

4) Write your R function to implement the coordinate descent algorithm from **2**). Fix the choice of the Lasso tuning parameter, $\lambda = 1.2\sqrt{\frac{\log(p)}{n}}$. Report your results including

- the number of iterations,
- the final iterate (only in the first 10 components), and
- plots of the iterates (only in the j th components, $j = 2, 4, 6, 8, 10$).

Note that an iteration is completed when all the components of u^0 are successively updated once. Check that the stationary condition from **1**) is properly satisfied.

5) Write your R function to implement the proximal gradient algorithm from **3**). Fix the choice of the Lasso tuning parameter, $\lambda = 1.2\sqrt{\frac{\log(p)}{n}}$, and use three choices of the step size, $\tau = [0.01, 0.005, 0.001]$. For each step size, report your results as in **4**), and check that the stationary condition from **1**) is properly satisfied.

Problem 2

This problem considers a real experiment conducted on 24 rats, 14 receiving a treatment, and 10 being in the control group. The measurements are taken on five sites (labeled as A-E) on the back of the rat from shoulder to tail. There are in total $n = 104$ measurements. The data are strictly positive strength measurements having a moderately large dynamic range from 3.8 to 82.5, so you will do a **log transformation** first. The data is stored in ‘‘qe_2024.txt’’ as a data frame, which you can load by `read.table(file="qe_2024.txt",header=T)`. The data is represented in

the wide format, so you will see some missing values (which is why there are only 104 measurements instead of $5 \times 24 = 120$). However, the analyses you will do are based on regression models, so there will not be any missing value issue. You will try to identify the **treatment effect** and **site effects** from the experimental data according to the instructions below. Let $1 \leq i \leq 24$ be the index of the rats, and $1 \leq s \leq 5$ be the index of the sites. For each rat i , let $t(i) = 1$ if it receives the treatment, and $t(i) = 0$ if it is in the control group. For this dataset, $t(i) = 1$ for $1 \leq i \leq 14$, and $t(i) = 0$ for $15 \leq i \leq 24$. We use y_{is} to denote the **log** measurement of the i -th rat at site s .

- (a) In the first regression model, assume

$$y_{is} = m_0 + \mu_{t(i)} + \epsilon_{is}, \quad \epsilon_{is} \stackrel{iid}{\sim} N(0, \sigma_0^2). \quad (2)$$

In other words, all measurements are viewed as independent normal random variables of the same variance σ_0^2 , with a common mean for the treatment group and another common mean for the control group. When you fit this regression model, use dummy variables for $\mu_{t(i)}$ with the constraint $\mu_0 = 0$. Report $\hat{\sigma}_0^2$. Perform the test $H_0 : \mu_1 = 0$ vs $H_1 : \mu_1 \neq 0$, and report the p -value.

- (b) Since the measurements are taken from five different sites of each rat, it is reasonable to consider the effect of the site, so the second regression model assumes

$$y_{is} = \beta_0 + \gamma_{t(i)} + \beta_s + \epsilon_{is}, \quad \epsilon_{is} \stackrel{iid}{\sim} N(0, \sigma^2). \quad (3)$$

Once again use dummy variables for $\gamma_{t(i)}$ with the constraint $\gamma_0 = 0$. Report the fitted model. Perform the test $H_0 : \gamma_1 = 0$ vs $H_1 : \gamma_1 \neq 0$, and report the p -value.

- (c) It is also reasonable to believe there could be correlations among the measurements from different sites on the same rat. The simplest way to introduce the correlation is to assume that the measurements on the same rat are equally correlated. Therefore, the third regression model assumes

$$y_{is} = \beta_0 + \gamma_{t(i)} + \beta_s + \epsilon_{is}, \quad (4)$$

where

$$\text{Cov}(\epsilon_{is}, \epsilon_{jr}) = \sigma_1^2 \delta_{ij} \delta_{sr} + \sigma_2^2 \delta_{ij}, \quad (5)$$

where δ_{ij} is the Kronecker symbol for equality of subscripts, and all ϵ_{is} are assumed to have a joint normal distribution. You will perform some derivation and analysis based on this model in (c)–(h).

Represent model (4) in the form

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}. \quad (6)$$

You should use dummy variables for $\gamma_{t(i)}$ and β_s , and set $\gamma_0 = 0$, $\beta_1 = 0$. So $\mathbf{X}$ is a $n \times 6$ matrix, and $\boldsymbol{\beta} = (\beta_0, \gamma_1, \beta_2, \beta_3, \beta_4, \beta_5)'$. Let s_i be the number of measurements available for the i -th rat. Order the elements of $\mathbf{y}$ by listing the s_1 measurement of the first rat, and then the s_2 measurements of the second rat, and so on and so forth. Report the design matrix $\mathbf{X}$.

- (d) We use k as the index of the vector $\mathbf{y}$ in (4), and define $i(k)$ as the rat index of the k -th element of $\mathbf{y}$. For example, $i(1) = \dots = i(4) = 1$, $i(5) = \dots = i(8) = 2$. Let $\mathbf{Z} = (z_{kj})$ be the $n \times 24$ matrix whose j -th column is defined as $z_{kj} = 1$ if $i(k) = j$ and 0 otherwise. For example,

$$\begin{pmatrix} z_{11} \\ z_{21} \\ z_{31} \\ z_{41} \\ z_{51} \\ z_{61} \\ z_{71} \\ z_{81} \\ z_{91} \\ \vdots \\ z_{n1} \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad \begin{pmatrix} z_{12} \\ z_{22} \\ z_{32} \\ z_{42} \\ z_{52} \\ z_{62} \\ z_{72} \\ z_{82} \\ z_{92} \\ \vdots \\ z_{n2} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 1 \\ 1 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

Show that $\text{Cov}(\boldsymbol{\epsilon}) = \sigma_1^2 \mathbf{I}_n + \sigma_2^2 \mathbf{Z} \mathbf{Z}'$, where σ_1^2 and σ_2^2 are defined in (5).

- (e) For any matrix, we use $\text{col}(\cdot)$ to denote the linear subspace spanned by its columns. Find an orthogonal matrix $\mathbf{U} = [\mathbf{U}_1, \mathbf{U}_2, \mathbf{U}_\perp]$ such that $\text{col}(\mathbf{U}_1) = \text{col}(\mathbf{X})$ and $\text{col}(\mathbf{U}_1, \mathbf{U}_2) = \text{col}(\mathbf{X}, \mathbf{Z})$. What are the dimensions of $\mathbf{U}_1$, $\mathbf{U}_2$ and $\mathbf{U}_\perp$?
- (f) Show that

$$\mathbf{U}_\perp' \mathbf{y} \sim N(\mathbf{0}, \sigma_1^2 \mathbf{I}).$$

Find the MLE of σ_1^2 based on $\mathbf{U}_\perp' \mathbf{y}$.

- (g) You have now found $\hat{\sigma}_1^2$ in (f). Take $\hat{\sigma}_2^2 = .148$ as granted, so $\text{Cov}(\boldsymbol{\epsilon})$ is estimated as

$$\widehat{\text{Cov}}(\boldsymbol{\epsilon}) = \hat{\sigma}_1^2 \mathbf{I}_n + \hat{\sigma}_2^2 \mathbf{Z} \mathbf{Z}'.$$

- (h) Now use $\widehat{\text{Cov}}(\boldsymbol{\epsilon})$ to fit the model (6) using generalized least squares. Report the fitted model. Perform the test $H_0 : \gamma_1 = 0$ vs $H_1 : \gamma_1 \neq 0$, and report the p -value.
- (i) Compare and comment on the three analyses.